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# Long-term renoprotection by perindopril or nifedipine in non-hypertensive patients with Type 2 diabetes and microalbuminuria

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## Abstract

**Aims** To assess the efficacy of an angiotensin converting enzyme (ACE) inhibitor (perindopril), a dihydropyridine calcium channel blocker (sustained release nifedipine) and placebo in preventing the progression of albuminuria and decline in glomerular filtration rate (GFR) in patients with Type 2 diabetes and microalbuminaria.

**Methods** A prospective, randomized, open, blinded end point study of 77 patients allocated to three treatment groups (23 perindopril, 27 nifedipine, 27 placebo). Drug doses were adjusted to achieve a decrease in diastolic blood pressure (DBP) of 5 mmHg in the first 3 months and additional therapy was given if hypertension developed (supine DBP > 90 mmHg and/or systolic blood pressure (SBP) > 140 mmHg if ≤ 40 years; supine DBP > 90 mmHg and/or SBP > 160 mmHg if > 40 years). Median follow-up was 66 months, with 37 patients being followed for at least 6 years.

**Results** Blood pressure remained within the non-hypertensive range in 83% of perindopril-, 95% of nifedipine- and 30% of placebo-treated patients ($P < 0.01$). In the first 12 months albumin excretion rate (AER) decreased by 47% only in the perindopril group ($P = 0.04$). From 12 to 72 months, AER gradients increased by 27% per year only in the placebo group ($P < 0.01$). After 6 years, macroalbuminuria had developed in 7/15 placebo compared with 2/11 in perindopril and 1/11 nifedipine-treated patients ($P = 0.05$). GFR did not change in the first 12 months, but thereafter the median GFR gradient (ml/min/1.73 $m^2$ per year) was −2.4 ($P < 0.01$) for perindopril-, −1.3 ($P = 0.26$) for nifedipine- and −4.2 ($P = 0.01$) for placebo-treated patients. The rate of decline in GFR for the study group as a whole from 12 months to the end of follow-up correlated negatively with mean arterial pressure (MAP) ($r = -0.38$, $P < 0.01$). During a 3-month treatment pause in 29 patients AER tended to increase only in the perindopril group ($P < 0.07$).

**Conclusions** Long-term control of blood pressure with perindopril or nifedipine stabilizes AER and attenuates GFR decline in proportion to MAP in non-hypertensive patients with Type 2 diabetes and microalbuminuria.

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**Keywords** microalbuminuria, nifedipine, non-hypertensive, perindopril, Type 2 diabetes

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**Abbreviations** ABCD, appropriate blood pressure control in Type 2 diabetes study; ACE, angiotensin converting enzyme; AER, albumin excretion rate; ANOVA, analysis of variance; DBP, diastolic blood pressure; GFR, glomerular filtration rate; $HbA_{1c}$, glycosylated haemoglobin $A_{1c}$; HPLC, high pressure liquid chromatography; MAP, mean arterial pressure; MDNSG, Melbourne Diabetic Nephropathy Study Group; PLSD, protected least significant difference; RAS, renin angiotensin system; SBP, systolic blood pressure

## Introduction

Patients with microalbuminuria and Type 2 diabetes have an increased risk of nephropathy and cardiovascular disease. Although most people with Type 2 diabetes and microalbuminuria have hypertension and blood pressure rises as albuminuria increases, three studies in normotensive microalbuminuric patients have shown that overt nephropathy develops in 23–50% over 5 years [1–3]. In both hypertensive and normotensive patients, treatment with an angiotensin converting enzyme (ACE) inhibitor [4], and in hypertensive patients treatment with an angiotensin receptor blocker [5], have been shown to lower or stabilize albumin excretion rate (AER) when compared with a placebo group with similar attained blood pressure. These agents have therefore been considered to possess renoprotective effects, independent of their anti-hypertensive action. Dihydropyridine calcium channel blockers are also effective anti-hypertensive agents, especially in the elderly [6], but it is not clear if they possess renoprotective effects.

In hypertensive patients with Type 2 diabetes and microalbuminuria, comparisons of ACE inhibitors and dihydropyridine calcium channel blockers have varied, with some showing a greater anti-albuminuric effect of ACE inhibition [7–9], and others showing no difference between the two therapies [10–12]. However, there have been no long-term comparisons of ACE inhibitors and dihydropyridine calcium channel blockers in normotensive Type 2 diabetic patients with microalbuminuria. Three long-term placebo controlled studies from Israel [1], Japan [8] and India [2], have shown that 4–5 years of treatment with an ACE inhibitor decreased or prevented an increase in AER, with only one study showing that ACE inhibitors preserved renal function as measured by the reciprocal of serum creatinine [1]. No similar placebo-controlled studies have been performed with calcium channel blockers.

A previous study by the Melbourne Diabetic Nephropathy Study Group (MDNSG) in normotensive patients with Type 1 diabetes and microalbuminuria showed that perindopril exerted a greater anti-albuminuric effect than sustained release nifedipine or placebo over 3 years [13]. Whether results from hypertensive patients can be extrapolated to the normotensive setting or whether results from normotensive patients with Type 1 diabetes can predict results in non-hypertensive Type 2 diabetic patients is not clear. The present study was therefore designed to compare the long-term effects of ACE inhibition with perindopril, dihydropyridine calcium channel blockade with sustained release (SR) nifedipine and placebo in patients with Type 2 diabetes, microalbuminuria and supine blood pressure less than 140 mmHg systolic and 90 mmHg diastolic. As in the two previous studies by the MDNSG [10,13], a treatment pause was added to delineate the long- and short-term effects of drug therapy.

## Methods

### Patients and methods

The MDNSG recruited patients from three University of Melbourne teaching hospitals: Austin Health, St Vincent's Hospital and The Western General Hospital, and one hospital from Monash University, the Monash Medical Centre. Ethics Committee approval was obtained from each centre and informed consent was obtained from all participants. This was a Prospective Randomized Open Blinded Endpoint evaluation study with participants being unaware of the identity of the study drug or the results of their laboratory investigations. Drug dispensing, laboratory measurements and statistical comparisons were performed independently of the treating doctors.

Sixty-nine Caucasian and eight Asian patients (seven Chinese, one Indian) were recruited into the study. The criteria for inclusion in the study were: patients aged 15–65 years, classified as having Type 2 diabetes for at least 1 year according to World Health Organization criteria [14], with microalbuminuria defined as at least two out of three consecutive measurements of AER between 20 and 200 μg/min, performed on overnight urine collections [15], and supine blood pressure less than 140 mmHg systolic and 90 mmHg diastolic, based on the median of three separate blood pressure measurements, two patients in the perindopril group had a median BP of 142/90 and 140/90 mmHg. All other patients had median BP levels less than 140/90 mmHg. Recruitment continued from 1992 to 1997. Exclusion criteria were any of the following: non-diabetic renal disease, serum creatinine ≥ 200 μM, haematuria, cardiac failure, hypertension, concurrent systemic disease, poor glycaemic control ($HbA_{1c}$ > 10%), serum potassium > 5 mM, recurrent urinary tract infection, patients at risk of pregnancy and patients with any other condition which might pose a risk to the patient or confound the results of the study.

The study was completed at the cessation of the pause period which ended in 2001. Of the 77 patients randomized, 37 patients had been followed for at least 72 months by the end of the study. The 40 patients who were followed less than 72 months included: eight patients who commenced the treatment pause before 72 months (due to the fact that patients were randomized over a period of years); 14 patients withdrawn due to

complications or death; five withdrawn due to relocation overseas or elsewhere; three patients withdrawn at their own request; three patients withdrawn after being enrolled in error; and seven patients who were lost to follow-up.

### Randomization and titration

The patients who met the above criteria were randomized to take either perindopril (Servier IRIS, Neuilly, France), nifedipine SR (Bayer, Wuppertal, Germany) or placebo therapy (in the image of a perindopril tablet, Servier IRIS, Neuilly, France). At each hospital, the randomization was performed by a pharmacist who had no information about individual patients, with the three treatments being randomized in blocks of six. The dose of placebo was one tablet daily. The initial dose of perindopril was 2 mg taken once daily in the morning and then titrated at 2 weekly intervals to 4 mg, then 8 mg taken once daily in the morning, aiming for a reduction in supine diastolic BP ≥ 5 mmHg. The initial dose of nifedipine SR was 10 mg twice daily, increasing to 20 mg twice daily, and then 40 mg twice daily if well tolerated, aiming for a reduction in supine diastolic BP ≥ 5 mmHg. Patients were seen 2 weekly or monthly until titration of their medication was completed, in all cases by 12 weeks after randomization.

### Endpoints

The major endpoint of the study was urinary AER. Secondary endpoints included the development of macroproteinuria or reversal to the normoalbuminuric range. If hypertension developed in the placebo group, defined as three consecutive monthly readings of supine diastolic blood pressure (DBP) > 90 mmHg, and/or supine systolic blood pressure (SBP) > 140 if aged ≤ 40 years, and/or supine SBP > 160 mmHg if aged > 40 years, or significant renal impairment was detected, defined as a glomerular filtration rate (GFR) < 60 ml/min/1.73 $m^2$ and/or serum creatinine > 0.20 mM, the study agent was ceased and the patient was allocated to either perindopril or nifedipine and dose titration occurred as outlined above. If hypertension persisted in the placebo group or developed in the other two groups, the patient was treated with hydrochlorothiazide 12.5–25 mg daily, atenolol 25–100 mg daily and then nifedipine or perindopril, depending on whether the patient was randomized to perindopril or nifedipine, respectively (Fig. 1).

### Study protocol

After randomization and titration of doses in the first 3 months all 77 participants were seen 3 monthly. There were 77 participants in the study, of whom 23 were randomized to perindopril, 27 to nifedipine and 27 to placebo therapy (Fig. 1). Six patients (two perindopril, two nifedipine, two placebo) were excluded from the study because of enrolment errors in four patients and loss to follow-up in two patients. Sixty-seven participants (18 perindopril, 24 nifedipine, 25 placebo) were followed for 12 months, 61 participants for 24 months and 50 patients for 48 months. Thirty-seven patients (11 perindopril, 11 nifedipine, 15 placebo) completed 72 months of follow-up.

The study was prematurely terminated by the investigators for several reasons. First, in the placebo group, about half of the subjects developed hypertension, requiring randomization to active treatment. Second, a number of seminal studies reported the benefits of blockade of the renin angiotensin system (RAS) in microalbuminuric Type 2 diabetic patients [4,5,16]. Third, the definition of hypertension in people with diabetes and the criteria for commencement of anti-hypertensive treatment and specifically blockade of the RAS changed, resulting in a significant proportion of placebo-treated subjects being eligible for blood pressure lowering therapy, including agents which interrupt the RAS [17,18].

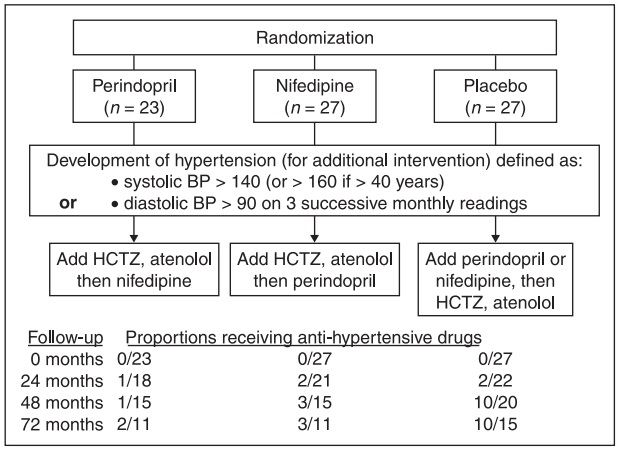


**Figure 1** Study design. BP, blood pressure; HCTZ, hydrochlorothiazide.

Every 3 months the following parameters were measured: supine clinic blood pressure after 5 min of rest, body weight, two consecutive measurements of overnight AER, $HbA_{1c}$, plasma renin concentration and urea and electrolytes. Every 12 months GFR was measured. At the end of the study a final GFR was measured in all patients in whom the interval between the termination date and the previous GFR was greater than 6 months. Serum cholesterol and triglycerides were measured every 12 months. At the end of the study, 29 patients ceased their randomized drug for 3 months and at the end of this treatment pause a final GFR and two consecutive overnight AER measurements were performed.

### Clinical methods

Office blood pressure was measured using an automated sphygmomanometer (Omron, model T4, HEM-739-C3) after 2 and 5 min in both supine and standing positions at each visit. Adjustments in therapy were based on the 5-min supine blood pressure. Fasting plasma cholesterol and triglycerides were measured by enzymatic colourimetric methods and plasma creatinine was measured by the modified Jaffe method using a Hitachi 747 autoanalyser (Roche Diagnostics, Mannheim, Germany). Urinary albumin excretion was measured by radioimmunoassay. Human albumin (Sigma, St Louis, MO, USA) was iodinated by the Chloramine T method. The detection limit of the assay was 15.6 ng/ml and the interassay coefficient of variation was 7.3% ($n$ = 37) at a concentration of 30 mg/l. All estimations of albumin concentration were performed at a single site on two urine collections per visit (Endocrinology Laboratory, Austin Health, Heidelberg, Australia). $HbA_{1c}$ was

measured by an automated HPLC technique (Biorad Diamat, Richmond, CA, USA). GFR was measured by a single injection technique using the isotope $99^m$ technetium diethylene triamine pentaacetic acid. GFR data were corrected for body surface area and the Brochner–Mortensen correction was applied [19]. Plasma renin concentration was measured by radioimmunoassay of angiotensin I generation [20]. Retinopathy was assessed on a dilated fundus by an ophthalmologist.

### Statistical methods

Continuous variables are expressed as mean ± SEM except where indicated. Differences in continuous variables were compared using analysis of variance (ANOVA) where subgroups were compared using Fisher's protected least significant difference (PLSD) post-hoc test. Differences in categorical variables were compared using chi square analysis (or Fisher's Exact test where expected cell frequencies were small). Time-to-event data for the preservation of normotension were compared using Kaplan–Meier survival methods and the log rank test. Gradients for AER, GFR and MAP were compared using non-parametric methods (Kruskal–Wallis test for between-group comparisons and the sign test for within-group comparisons). Logarithmic transformation of AER was performed prior to statistical analysis. All analyses were performed on an intention-to-treat basis. A level of $P < 0.05$ was considered statistically significant and 95% levels of confidence were used for confidence intervals. Statistical analysis was performed using Stata (StataCorp, College Station, TX, USA, 1997) and Stat View (SAS Institute, Cary, NC, USA, 1998). At the end of the study the effects of 3 months' withdrawal of randomized therapy were assessed by analysis of variance.

## Results

The mean age of participants was 53 ± 1 years. There were no baseline differences among the three groups in gender ratio, diabetes duration, blood pressure, AER, glycaemic control, lipid profile, weight, smoking habits or presence of retinopathy (Table 1). Baseline plasma creatinine was unevenly distributed but the three groups were well matched for GFR. Baseline geometric mean AER was: perindopril 59, nifedipine 55 and placebo 62 μg/min. MAP at baseline was perindopril 100 ± 2, nifedipine 99 ± 1 and placebo 99 ± 1 mmHg and group means for MAP were maintained below 100 mmHg throughout the 6 year follow-up in the perindopril and nifedipine groups.

The mean $HbA_{1c}$ during the study was similar in the three study groups: perindopril 8.4 ± 0.3%, nifedipine 8.2 ± 0.2% and placebo 8.5 ± 0.3%. The plasma renin concentration was increased in the perindopril group at 36 months (4.6 ± 1.2 μg/ml/h) compared with nifedipine (2.1 ± 0.4 μg/ml/h, $P < 0.05$) and placebo (2.0 ± 0.5 μg/ml/h, $P < 0.02$). Mean serum cholesterol levels remained similar over 72 months in the three groups: perindopril 5.9 ± 0.2 mM, nifedipine 5.4 ± 0.1 mM and placebo 5.8 ± 0.2 mM. Mean serum triglyceride levels over 72 months were also similar: perindopril 2.5 ± 0.2 mM, nifedipine 2.0 ± 0.2 mM and placebo 2.4 ± 0.2.

Twenty of the 77 patients developed hypertension during the study, with the majority being in the placebo group (perindopril 3/23, nifedipine 4/27, placebo 13/27, $P < 0.01$). At 6 years of follow-up, time-to-event analysis showed that hypertension was avoided in 83% (95% CI, 62–100%) of

**Table 1** Baseline characteristics of the three treatment groups

| Variable | Perindopril ($n$ = 23) | Nifedipine ($n$ = 27) | Placebo ($n$ = 27) | $P$ |
|---|---|---|---|---|
| Age (years) | 50 ± 2 | 55 ± 2 | 53 ± 1 | 0.25 |
| Sex (M/F) | 14/9 | 20/7 | 15/12 | 0.35 |
| Diabetes duration (years) | 5 ± 1 | 9 ± 1 | 9 ± 2 | 0.17 |
| AER (μg/min) | 59 ×/÷ 1.1 | 55 ×/÷ 1.1 | 62 ×/÷ 1.1 | 0.76 |
| bmGFR† (ml/min/1.73 $m^2$) | 92 ± 8 | 87 ± 5 | 98 ± 6 | 0.40 |
| Plasma creatinine (μM) | 71 ± 4 | 87 ± 4 | 79 ± 2 | 0.02 |
| Systolic BP (mmHg) | 139 ± 3 | 137 ± 2 | 136 ± 3 | 0.66 |
| Diastolic BP (mmHg) | 81 ± 1 | 81 ± 1 | 81 ± 1 | 0.97 |
| Mean arterial BP (mmHg) | 100 ± 2 | 99 ± 1 | 99 ± 1 | 0.72 |
| $HbA_{1c}$ (%) | 8.0 ± 0.4 | 8.1 ± 0.4 | 8.5 ± 0.3 | 0.61 |
| Cholesterol (mM) | 6.0 ± 0.3 | 5.5 ± 0.2 | 6.3 ± 0.4 | 0.20 |
| Triglycerides (mM) | 2.4 ± 0.2 | 2.8 ± 0.6 | 3.3 + 0.5 | 0.47 |
| Weight (kg) | 85 ± 3 | 84 ± 4 | 84 ± 4 | 0.96 |
| Smoking (%) | 25% | 38% | 28% | 0.57 |
| Retinopathy (%) | 20% | 38% | 36% | 0.37 |
| Diet/oral/insulin (%) | 11, 47, 42 | 0, 68, 32 | 13, 43, 42 | 0.33 |
| Lipid treatment (%) | 59 | 29 | 64 | 0.08 |
| Plasma renin concentration 0.25 (ng/ml/h) | 0.9 ×/÷ 1.2 | 1.1 ×/÷ 1.3 | 1.6 ×/÷ 1.2 | |

Continuous data are mean ± SEM except for geometric mean ×/÷ tolerance factor for AER and plasma renin concentration.
$P$-values refers to one-way ANOVA for continuous variables, chi square test for categorical variables.
†Brochner-Mortensen corrected DTPA-GFR.

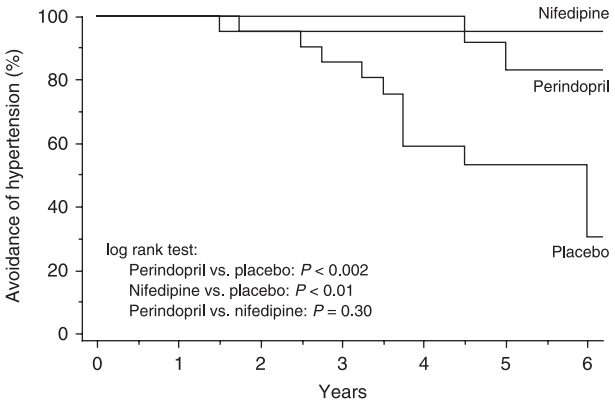


**Figure 2** Time-to-event analysis for the preservation of normotension. At 6 years, normotension was preserved in 83% of perindopril, 95% of nifedipine and 30% of placebo treated patients.

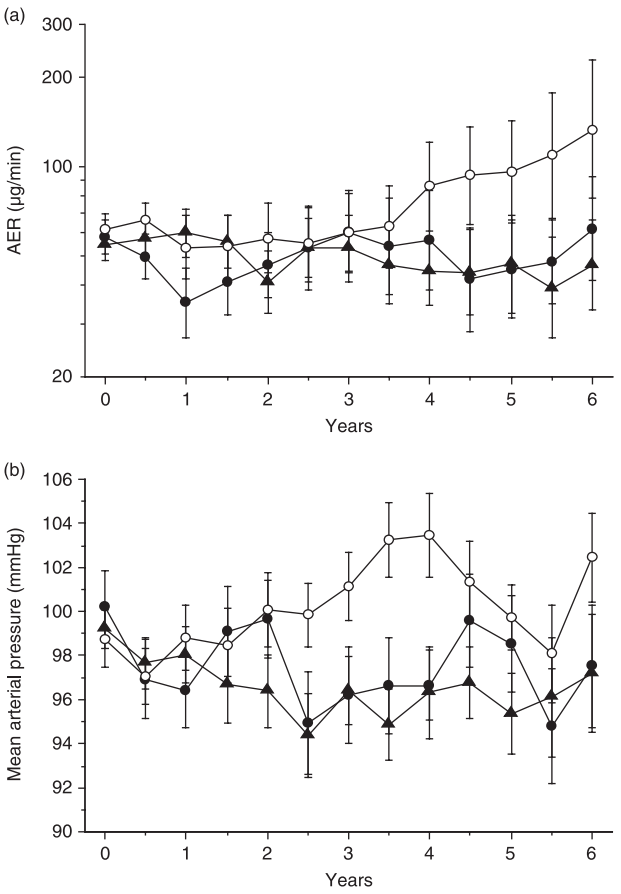


**Figure 3** Data over a 6-year treatment period for (a) AER and (b) mean arterial pressure. Data shown as geometric mean ×/÷ tolerance factor on a logarithmic scale for AER, and mean ± SEM for MAP. (●) perindopril, (▲) nifedipine and (○) placebo.

perindopril-treated, 95% (86–100%) of nifedipine-treated in comparison with only 30% (7–54%) of placebo-treated patients ($P < 0.01$) (Fig. 2). The proportions of patients receiving anti-hypertensive drugs other than allocated study therapy at 24, 48 and 72 months, respectively, were 1/18, 0/15 and 2/11 in the perindopril group, 2/21, 3/15 and 3/11 in the nifedipine group and 2/22, 10/20 and 10/15 in the placebo group, respectively. MAP was similar in all three groups for the first 24 months (Fig. 3) but was significantly lower in the perindopril ($P < 0.01$) and nifedipine group ($P < 0.01$) from 30 to 48 months compared with the placebo group (repeated measures ANOVA, $P < 0.01$). At 48 months the MAP for each group was perindopril 97, nifedipine 96, and placebo 103 mmHg ($P < 0.02$).

AER gradients in the three groups showed a biphasic response (Fig. 3). From baseline to 12 months the AER gradient showed a decreasing trend in the perindopril group (median change –47% per year, $P = 0.12$) which was greater than that observed in the other two groups (nifedipine +17% per year, placebo –10% per year, $P = 0.04$). By contrast, from 12 to 72 months the AER gradient increased significantly within the placebo group (+28% per year, $P < 0.01$) while remaining stable in the perindopril (+2% per year) and nifedipine (+4% per year) groups. Analysis of AER at 72 months revealed similar rates of reversal to normoalbuminuria, 1/11 (9%) in the perindopril group, 3/11 (27%) in the nifedipine group and 3/15 (20%) in the placebo group. In the placebo group, reversal to normoalbuminuria occurred spontaneously in two patients and after commencement of anti-hypertensive therapy in one patient. Macroalbuminuria developed in 2/11 (18%) of the perindopril group, 1/11 (9%) of the nifedipine group and 7/15 (47%) of the placebo group ($P = 0.05$).

Baseline GFR was similar in the three groups: perindopril 92 ± 8, nifedipine 87 ± 5 and placebo 98 ± 6 ml/min/1.73 $m^2$. Median GFR gradients were assessed from 0 to 12 months and from 12 months to the final yearly measurement. From baseline to 12 months, there was no significant difference in the median GFR gradient in the three groups. In 58 patients with minimum follow-up of 24 months and median total follow-up of 66 months (perindopril 66, nifedipine 54 and placebo 69 months) the median GFR gradient (ml/min/1.73 $m^2$ per year) from 12 months to final follow-up was –2.4 ($P < 0.01$) in the perindopril group, –1.3 ($P = 0.26$) within the nifedipine group and –4.2 ($P = 0.01$) within the placebo group. There was a trend for a greater rate of decline in GFR in the placebo group which just failed to reach statistical significance (group comparison, $P = 0.056$, placebo vs. nifedipine and perindopril groups). The effect of achieved MAP on rate of decline in GFR was examined in the total cohort. After excluding GFR data from the first 12 months of treatment, there was a significant negative correlation between rate of decline in GFR and achieved MAP from 12 months to the end of follow-up (Fig. 4).

After a median study duration of 81 months, randomized study treatment was ceased for 3 months in all 29 eligible subjects (perindopril 8, nifedipine 10, placebo 11). During the treatment pause there were no significant changes in MAP, AER or GFR among the three groups, although there was a trend for an increase in AER in the perindopril group (Table 2).

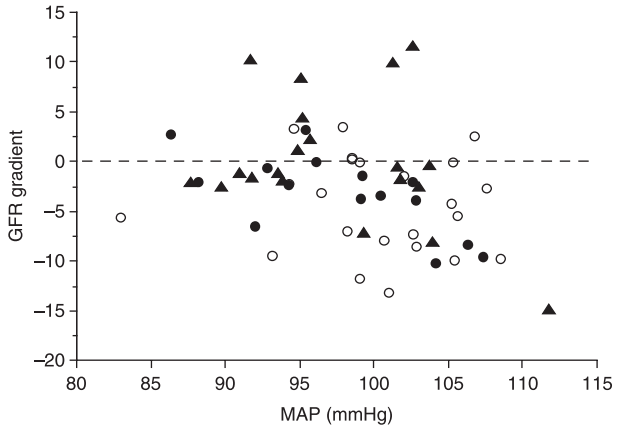


**Figure 4** Relationship between the rate of change in GFR, expressed as a gradient (ml/min/1.73 $m^2$ per year) from 12 months to final follow-up, and the mean arterial pressure during the same period in all study patients (Spearman's rho = –0.38, $P < 0.01$). (●) perindopril, (▲) nifedipine and (○) placebo.

## Discussion

The main finding in this study was that long-term treatment with perindopril or nifedipine prevented an increase in blood pressure and stabilized AER in non-hypertensive patients with Type 2 diabetes and microalbuminuria. Compared with placebo-treated patients, perindopril and nifedipine both lowered MAP by 5–7 mmHg from 30 to 48 months and stabilized albumin excretion rate to a similar degree over 6 years. In the previous MDNSG study in normotensive Type 1 diabetic patients with microalbuminuria, perindopril treatment resulted in a 45% decrease in median albumin excretion rate in the first year [13], which is almost identical to the 47% decrease observed in the present study. After the first year in the Type 1 diabetes study, albumin excretion rate stabilized in the perindopril group, but rose progressively over the next 2 years in the nifedipine and placebo groups [13]. By contrast, the anti-albuminuric effects of perindopril and nifedipine were comparable in the present study.

The results of this study are similar to those of the normotensive ABCD (appropriate blood pressure control in Type 2 diabetes) study which compared the renal effects of ACE inhibition with enalapril and calcium channel blockade with nisoldipine in 480 normotensive Type 2 diabetic patients, 23% of whom had microalbuminuria [3]. No difference was found between the use of enalapril or nisoldipine on the progression of albuminuria in the normoalbuminuric or microalbuminuric subgroups. In a subgroup of 109 patients with baseline microalbuminuria, intensive blood pressure control reduced the rate of progression of AER and the development of macroalbuminuria. The number of participants with microalbuminuria in the intensive and moderate blood pressure control groups of the ABCD study was comparable with the present study. The results of both studies indicate that prevention of hypertension is an important determinant of progression of AER in normotensive patients with Type 2 diabetes and microalbuminuria.

The lack of a superior effect of perindopril on AER compared with nifedipine in the present study may have several explanations. First, it is possible that this study, with a relatively small sample size, did not have sufficient statistical power. Nevertheless, the previous MDNSG study in a smaller group of normotensive patients with Type 1 diabetes and microalbuminuria using an identical protocol did reveal a significantly greater anti-albuminuric effect of perindopril compared with nifedipine or placebo over 3 years [13]. Second, it may be that the specific renoprotective effects of ACE inhibitors, compared with other anti-hypertensive agents, are attenuated at lower levels of blood pressure. This has been suggested in a meta-analysis of studies of the effects of ACE inhibition in patients with overt nephropathy [21] and has also been proposed by Mogensen [22]. Third, non-hypertensive patients with Type 2 diabetes and microalbuminuria may represent a qualitatively different cohort when compared with hypertensive patients with microalbuminuria.

It is also possible that variations in the disease process may underlie the different responses to anti-hypertensive therapy in non-hypertensive, microalbuminuric patients with Type 2 diabetes compared with responses in Type 1 diabetes. These

**Table 2** Data shown for AER, GFR and MAP performed on the last month ON treatment and then OFF treatment after a 3-month pause in 29 eligible patients. (withdrawal of randomized drug only)

| Variable | Perindopril ($n$ = 8) | Nifedipine ($n$ = 10) | Placebo* ($n$ = 11) | $P$ (between group) |
|---|---|---|---|---|
| AER ON (final) | 52 ×/÷ 1.3 | 58 ×/÷ 1.7 | 42 ×/÷ 1.5 | 0.86 |
| AER OFF (3 months) | 80 ×/÷ 1.6 | 78 ×/÷ 1.6 | 48 ×/÷ 1.6 | 0.70 |
| $P$ (within group) | < 0.07 | 0.13 | 0.60 | |
| bmGFR ON (final) | 87 ± 6 | 89 ± 10 | 78 ± 9 | 0.65 |
| bmGFR OFF (3 months) | 78 ± 7 | 84 ± 9 | 74 ± 8 | 0.68 |
| $P$ (within group) | < 0.09 | 0.39 | 0.20 | |
| MAP ON (final) | 99 ± 3 | 101 ± 2 | 96 ± 2 | 0.34 |
| MAP OFF (3 months) | 98 ± 3 | 101 ± 3 | 99 ± 3 | 0.82 |
| $P$ (within group) | 0.64 | 0.90 | 0.34 | |

Data shown as geometric mean ×/÷ tolerance factor for AER, and as mean ± SEM for GFR and MAP. $P$-values refer to ANOVA (between groups) and paired $t$-tests (within groups).
*8/11 placebo patients were on anti-hypertensive therapy which continued during the treatment pause.

include the effects of age [23], the level of blood pressure, even within the 'normotensive' range and ethnicity [24]. In the present study involving predominantly Caucasian patients, it was confirmed that ACE inhibition is superior to placebo in non-hypertensive, microalbuminuric patients, as has been previously reported in Asian patients [8].

It has previously been postulated by Bakris *et al.*, based on a meta-analysis, that the rate of decline in GFR in patients with overt nephropathy is related to the attained MAP [25]. The present study extends this concept to the microalbuminuric Type 2 diabetic population. Specifically, it was shown that the rate of decline in GFR, from 12 months to the end of follow-up, was related to mean arterial pressure. Indeed, GFR appeared to be stabilized once MAP was < 95 mmHg independent of the type of anti-hypertensive agent used. A previous meta-analysis by Weidmann also suggested that once achieved blood pressure is low, no superiority of any anti-hypertensive agent can be detected [21].

In contrast to the clear rise in AER previously observed during treatment pauses after 1 year [10], or after 3 years [13], a 3-month cessation of the originally allocated study drug at 81 months in the present study was not followed by a significant rebound of AER in either the perindopril (trend $P < 0.07$) or nifedipine groups. This suggests that AER during long-term anti-hypertensive therapy no longer reflects functional alterations in the glomerular filter, renal tubules or renal vasculature.

In the present study, 10/15 patients in the placebo group had progressed to active therapy at 72 months. The study protocol, though considered appropriate ethically when the study started approximately 10 years ago, demanded active therapy in the placebo group once blood pressure started to rise. This may have decreased the chances of detecting a difference between the active and placebo groups. Though not our original intention, the question we addressed was whether anti-hypertensive therapy should be commenced in non-hypertensive patients with microalbuminuria or whether it can be delayed until hypertension develops. This study has answered this question for an older definition of hypertension [26], but the definition of hypertension has now changed, with equal recognition of systolic hypertension as well as diastolic hypertension especially in the context of diabetes [17,18]. Indeed, our data provide justification for the changed definition of hypertension in diabetes, as some participants in the present study, who were categorized as 'normotensive' as recently as 1996 would now be considered hypertensive [18,25,27]. Independent of the type of drug used, these patients achieved blood pressure targets that are now recommended. This study therefore provides evidence to justify the new targets, as patients treated with perindopril or nifedipine did not progress in terms of albuminuria over 6 years and the group means for rate of decline of GFR were close to that expected from age alone [23,28].

The results of the present study suggest that small increases in blood pressure, similar to those reported in Type 1 diabetes [29], may have a pathogenetic role in the transition from normo- to microalbuminuria and in the progression of microalbuminuria in Type 2 diabetes. Larger prospective randomized studies are therefore required to elucidate fully whether genuinely normotensive Type 2 diabetic patients with microalbuminuria should be on anti-hypertensive therapy.

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